Week 4 lecture: second-order methods

Newton’s method uses curvature as well as slope. Each step solves H(x) d = -grad f(x) for the direction d and moves to x + d, where H is the Hessian matrix of second derivatives.

Near a minimum with a positive definite Hessian, Newton’s method converges quadratically: the number of correct digits roughly doubles every step.

The cost of curvature

Forming the Hessian takes n^2 entries and solving the linear system takes about n^3 operations, which is too expensive when a model has millions of parameters.

Quasi-Newton methods such as BFGS build an approximation of the inverse Hessian from successive gradient differences. L-BFGS keeps only the last few difference pairs, so memory grows linearly in n.

Line search

Instead of a fixed step, a backtracking line search starts with a full step and halves it until the Armijo condition holds:
f(x + t d) <= f(x) + c t grad f(x)^T d, with c around 1e-4.
This guarantees enough decrease without tuning a learning rate.